2 Parallel or accent-controlled corpus design

Saturday, August 1, 2026 6:55 PM

Accent conversion using discrete units with parallel data synthesized from controllable accented TTS

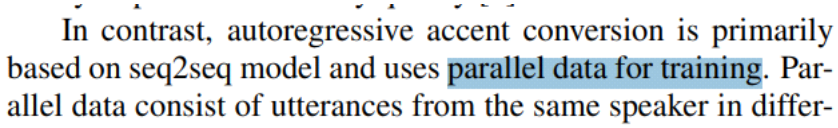
Title: Accent conversion using discrete units with parallel data synthesized from controllable accented TTS

Year: 2024

BibTeX Citation:

```
@unknown{unknown,
author = {Nguyen, Tuan and Pham, Ngoc and Waibel, Alexander},
year = {2024},
month = {09},
pages = {},
title = {Accent conversion using discrete units with parallel data synthesized from controllable accented TTS},
doi = {10.48550/arXiv.2410.03734}
}
```

Was it parallel (fixed script) or not?: YES



In contrast, autoregressive accent conversion is primarily based on seq2seq model and uses parallel data for training. Parallel data consist of utterances from the same speaker in differ-

What the parallel/controlled design enabled (the borrowed justification):

The parallel design enabled the researchers to train a sequence-to-sequence model that converts multiple non-native accents into a native accent without needing reference audio during inference (zero-shot, reference-free), while actively improving speaker fluency

Language: English

Domain: Artificial Intelligence, Deep Learning, Speech Synthesis, and Accent Conversion (AC)

Why parallelism is necessary:

Parallelism is necessary because non-parallel (non-autoregressive) methods fail to improve the actual fluency and pacing of the non-native speaker, as they cannot alter the duration of the input audio. Parallel data allows the model to learn and correct these timing and fluency errors

Additional attributes:

Data Augmentation / Parallel Synthesis Strategy:

The authors designed a unique method to overcome the scarcity of parallel data by synthetically generating it. They tested two specific strategies: non-overlapped sentences and overlapped sentences.

In our research, we also investigate how data augmentation can affect the performance of the PC. In details, we devise two ways for synthesizing 1 million utterances. We can select one million sentences from a text corpus and generate one non-native audio recording of each sentence with a random accent and speaker, which we call non-overlapped sentence strategy. The second strategy involves selecting 166 thousand sentences from a text corpus and creating non-native 6 audio files per sentence, each sentence with all 6 accents and 6 random speakers, which we call overlapped sentence strategy. In both circumstances, we generate one native audio per sentence as output audio. The second way, we believe, could help PC learn how to address varied accents more successfully. Before generating synthetic data, we divided the sentences into train and validation sets in the ratio 1000:1. The test data is our in-house data, which contains approximately 1000 sentences (3 hours) recorded by Chinese, Indian, Arabic and Vietnamese speakers. These speakers have not been recorded in training data, so we can consider as zero-shot condition.

Model Architecture (The Pipeline):

cascaded architecture featuring a sequence-to-sequence (seq2seq) model and a Unit-to-Speech (U2S) model

Corpora/Datasets Used for Training:

The study relies on several established datasets to build their synthetic parallel corpus, including LJSpeech, LibriTTS-R, L2-Arctic, and Common Voice.

An international English speech corpus for longitudinal study of accent development

Title: An international English speech corpus for longitudinal study of accent development

Year: 2011

BibTeX Citation:

```
@inproceedings{inproceedings,
author = {Orr, Rosemary and Quené, Hugo and Beek, Roeland and Diefenbach, Thari and Van Leeuwen, David and Huijbregts, Marijn},
year = {2011},
month = {08},
pages = {1889-1892},
title = {An International English Speech Corpus for Longitudinal Study of Accent Development.},
journal = {Scandinavian Journal of Immunology - SCAND J IMMUNOL},
doi = {10.21437/Interspeech.2011-513}
}
```

Was it parallel (fixed script) or not?:

Yes, it relies heavily on parallel, read speech combined with conversational speech

What the parallel/controlled design enabled (the borrowed justification):

A fixed script allowed researchers to use semi-automatic scripts to rapidly isolate target sentences and run forced-aligned phone recognition to measure accent convergence over time.

Language:

English (specifically used as a *lingua franca* among international speakers).

Domain:

Sociolinguistics, Longitudinal Accent Development, Speech Intelligibility, and Speech Perception

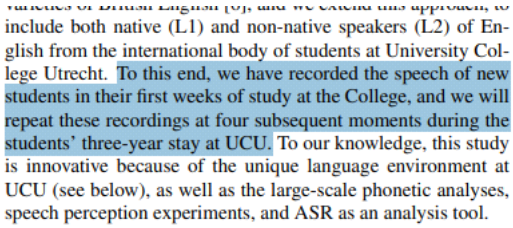
Why parallelism is necessary:

Parallelism is necessary to control acoustic variables (like intensity) so that changes in intelligibility and phonetic production can be accurately mapped to the speaker's accent evolution rather than varying text.

Additional attributes:

Data Augmentation / Parallel Synthesis Strategy:

this study does not use data augmentation or synthetic speech generation. Instead, it focuses on capturing natural human recordings repeatedly over a five-year period to track real-world accent evolution


varieties of British English [6], and we extend this approach, to include both native (L1) and non-native speakers (L2) of English from the international body of students at University College Utrecht. To this end, we have recorded the speech of new students in their first weeks of study at the College, and we will repeat these recordings at four subsequent moments during the students' three-year stay at UCU. To our knowledge, this study is innovative because of the unique language environment at UCU (see below), as well as the large-scale phonetic analyses, speech perception experiments, and ASR as an analysis tool.

Model Architecture (The Pipeline):

Corpora/Datasets Used for Training:

Evaluation Metrics:

The primary purpose of this paper is the creation of a new dataset (the UCU longitudinal corpus) rather than training a model on existing ones. However, they do note the dataset used to train their baseline automatic speech recognition (ASR) tool for forced alignment

---

Analyzing the Impact of Accent on English Speech: Acoustic and Articulatory Perspectives

Title: Analyzing the Impact of Accent on English Speech: Acoustic and Articulatory Perspectives

Year: 2025

BibTeX Citation:

```
@inproceedings{Premananth_2025, series={interspeech_2025},
   title={Analyzing the Impact of Accent on English Speech: Acoustic and Articulatory
Perspectives},
   url={http://dx.doi.org/10.21437/Interspeech.2025-2342},
   DOI={10.21437/interspeech.2025-2342},
   booktitle={Interspeech 2025},
   publisher={ISCA},
   author={Premananth, Gowtham and Kugathasan, Vinith and Espy-Wilson, Carol},
   year={2025},
   month=Aug, pages={1493–1497},
   collection={interspeech_2025} }
```

Was it parallel (fixed script) or not?:

Yes, it relies exclusively on parallel datasets where speakers read the exact same

prompts or passages.

CMU ARCTIC Dataset [13]: This corpus comprises recordings from four native English speakers (two male, two female) sampled at 16 kHz. Each speaker reads approximately 1,300 phonetically balanced English prompts (ARCTIC prompts). In this study, we use this dataset as the non-accented baseline.

What the parallel/controlled design enabled (the borrowed justification):

The parallel design enabled the researchers to average their acoustic and articulatory results across phonetically identical utterances, effectively removing individual phonetic biases and allowing for a direct, "apples-to-apples" comparison of accent coordination.

Language: English

Domain: Speech processing, Acoustic and Articulatory Analysis.

Why parallelism is necessary:

Accented speech varies significantly depending on the specific accent, influencing English pronunciation in unique ways. As a result, a generalized analysis of how accents affect speech is insufficient. Instead, it is crucial to deeply understand the distinct impacts of individual accents. While prior studies [11, 12] have attempted to measure the strength of foreign accents in

Additional attributes:

Data Augmentation / Parallel Synthesis Strategy:

this study does not synthesize or augment data. It strictly extracts acoustic and articulatory features from real human speech found within existing parallel corpora

Model Architecture (The Pipeline):

Their analytical pipeline involves extracting low-level features (like Pitch, MFCCs, and Vocal Tract Variables via a deep learning Speech Inversion system) and converting them into high-level channel-delay correlation matrices (eigenspectra) to measure coordination

Corpora/Datasets Used for Training:

study analyzes three specific, widely-used parallel datasets.

L2-ARCTIC: A Non-Native English Speech Corpus

Title: L2-ARCTIC: A Non-Native English Speech Corpus

Year: 2018

BibTeX Citation:

```
@inproceedings{inproceedings,
author = {Zhao, Guanlong and Sonsaat, Sinem and Silpachai, Alif and Lucic, Ivana and
Chukharev-Hudilainen, Evgeny and Levis, John and Gutierrez-Osuna, Ricardo},
year = {2018},
month = {09},
pages = {2783-2787},
title = {L2-ARCTIC: A Non-native English Speech Corpus},
doi = {10.21437/Interspeech.2018-1110}
}
```

Was it parallel (fixed script) or not?:

Yes, it relies entirely on a heavily controlled, parallel script.

What the parallel/controlled design enabled (the borrowed justification):

The controlled, parallel design enabled the researchers to generate forced-aligned phonetic transcriptions and directly evaluate systems designed to map a non-native speaker's accent onto a native target voice (or vice versa).

Language:

English (spoken by non-native speakers of Hindi, Korean, Mandarin, Spanish, and Arabic).

Domain:

Voice conversion, accent conversion, and computer-assisted pronunciation training / mispronunciation detection

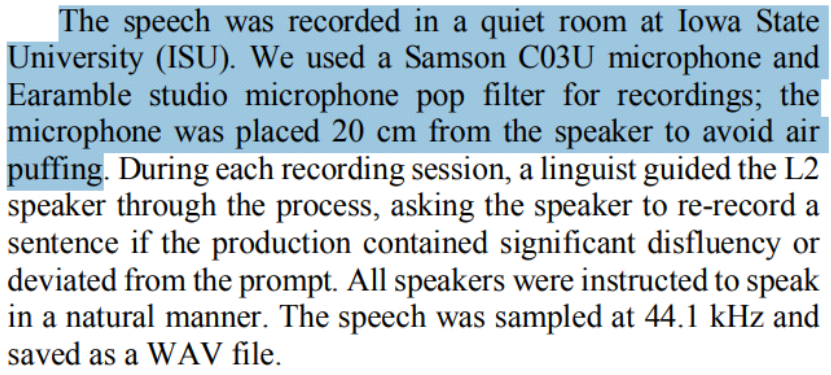
Why parallelism is necessary:

Parallelism is necessary because without high-quality recordings of different speakers reading the exact same linguistic content, it is nearly impossible to train algorithms to perform voice and accent conversion. Existing non-parallel corpora were too noisy or lacked the necessary structure.

Additional attributes:

Data Augmentation / Parallel Synthesis Strategy:

this paper introduces a recorded human corpus rather than a synthetic model, no data augmentation or synthesis was used to create the core dataset. Instead, they utilized controlled studio recording procedures.



The speech was recorded in a quiet room at Iowa State University (ISU). We used a Samson C03U microphone and Earamble studio microphone pop filter for recordings; the microphone was placed 20 cm from the speaker to avoid air puffing. During each recording session, a linguist guided the L2 speaker through the process, asking the speaker to re-record a sentence if the production contained significant disfluency or deviated from the prompt. All speakers were instructed to speak in a natural manner. The speech was sampled at 44.1 kHz and saved as a WAV file.

Model Architecture (The Pipeline):

To process the raw audio into a usable dataset, their pipeline involved automatic forced-alignment followed by rigorous human annotation. They also established a baseline mispronunciation detection model using the Goodness of Pronunciation (GOP) method based on a GMM acoustic model.

Corpora/Datasets Used for Training:

The researchers used the text prompts from an existing parallel corpus (CMU ARCTIC) to generate the script for their new corpus (L2-ARCTIC). For evaluating their baseline mispronunciation detection system, they utilized an acoustic model trained on the Librispeech corpus.

Open-source Multi-speaker Corpora of the English Accents in the British Isles

Title: Open-source Multi-speaker Corpora of the English Accents in the British Isles

Year: 2020

BibTeX Citation:

@inproceedings{inproceedings,

```
author = {Demirsahin, Isin and Kjartansson, Oddur and Gutkin, Alexander and Rivera, Clara},
year = {2020},
month = {05},
pages = {},
title = {Opensource Multispeaker Corpora of the English Accents in the British Isles}
}
```

Was it parallel (fixed script) or not?:

Yes, the dataset heavily relies on a highly curated, fixed parallel script containing both global and regional lines.

This paper presents a dataset of transcribed high-quality audio of English sentences recorded by volunteers speaking with different accents of the British Isles. The dataset is intended for linguistic analysis as well as use for speech technologies. The recording scripts were curated specifically for accent elicitation, covering a variety of phonological phenomena and providing a high phoneme coverage. The scripts include pronunciations of global locations, major airlines and common personal names in different accents; and native speaker pronunciations of local words. Overlapping lines for all speakers were included for idiolect elicitation, which include the same or similar lines with other existing resources such as the CSTR VCTK corpus and the Speech Accent Archive to allow for easy comparison of personal and regional accents. The resulting corpora include over 31 hours of recordings from 120 volunteers who self-identify as native speakers of Southern England, Midlands, Northern England, Welsh, Scottish and Irish varieties of English.

What the parallel/controlled design enabled (the borrowed justification):

The controlled design enabled the researchers to isolate a speaker's unique personal quirks from broader regional dialect traits, and also allowed for direct cross-referencing with other major datasets

Language:

English (specifically targeting Southern England, Midlands, Northern England, Welsh, Scottish and Irish varieties).

Domain:

Linguistic analysis, phonology, and speech technologies (like Text-to-Speech).

Why parallelism is necessary:

Parallelism (using identical target words) is necessary because natural language evolution causes phonemes to merge or split unpredictably; without forcing speakers to say the exact same curated words (like "meet" vs. "meat"), you cannot definitively map out a dialect's specific phonological rules.

Additional attributes:

Data Augmentation / Parallel Synthesis Strategy:

This study introduces a human-recorded corpus, so no synthetic generation or data augmentation was used. Instead, they relied on structured crowd-sourcing and volunteer recruitment in controlled environments.

Model Architecture (The Pipeline):

Because they are building a dataset rather than an AI model, their "pipeline" consists of rigorous acoustic hardware standardization to ensure the data is high enough quality for speech synthesis

Corpora/Datasets Used for Training:

While they did not train a model, they explicitly embedded text from other major parallel corpora (like VCTK and the Speech Accent Archive) into their own recording scripts so that their new data could be directly compared to older datasets

---

Recognition of Arabic Accents From English Spoken Speech Using Deep Learning Approach

Title: Recognition of Arabic Accents From English Spoken Speech Using Deep Learning Approach

Year: 2024

BibTeX Citation:

```
@article{article,
author = {Habbash, Mansoor and Sami, Mnasri and Alghamdi, Mansoor and Alrashidi, Malek and Tarawneh, Ahmad and Gumair, Abdullah and Hassanat, Ahmad},
year = {2024},
month = {01},
pages = {37219-37230},
title = {Recognition of Arabic Accents From English Spoken Speech Using Deep Learning Approach},
volume = {12},
journal = {IEEE Access},
doi = {10.1109/ACCESS.2024.3374768}
}
```

Was it parallel (fixed script) or not?:

Yes, it relies completely on a fixed, parallel script read by all participants.

What the parallel/controlled design enabled (the borrowed justification):

The controlled text enabled the researchers to test machine learning algorithms' ability to distinguish between distinct regional accents by isolating the acoustic differences in how specific, identical words are pronounced across different countries.

Language: English (spoken by non-native Arabic speakers from Jordan, Iraq, Saudi Arabia, and Tunisia).

Domain: Deep learning, accent recognition, automatic speech recognition (ASR), and classification.

Why parallelism is necessary:

Parallelism (using a standardized script like the "Stella" passage) is necessary to ensure experimental comparability with other major linguistic studies on human-accented speech.

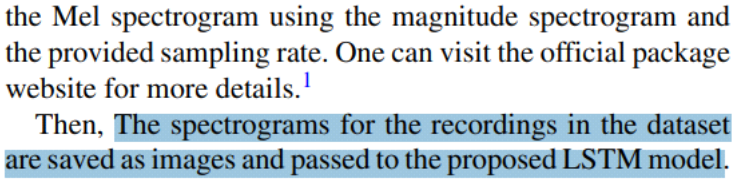
Additional attributes:

Data Augmentation / Parallel Synthesis Strategy:

This study did not use synthetic data augmentation. Instead, they focused on preprocessing and unifying the raw human audio recordings by standardizing the sampling rate and file formats for clean feature extraction.

Model Architecture (The Pipeline):

The pipeline relies on converting the audio into Mel spectrograms, treating those spectrograms as images, and passing them through a Long Short-Term Memory (LSTM) neural network. They also establish baselines by running the data through MLP, 1-DCNN, and BiLSTM models.



the Mel spectrogram using the magnitude spectrogram and the provided sampling rate. One can visit the official package website for more details.[1]

Then, The spectrograms for the recordings in the dataset are saved as images and passed to the proposed LSTM model.

Corpora/Datasets Used for Training:

Because they found existing online datasets lacked sufficient Arabic recordings for deep learning, they curated a brand new, in-house dataset consisting of 391 recordings.
